Problem Sheet -1.

3/Aug/2020

1. a. Correct - 6 is the highest order
   b. Correct - 3 is the highest order
   c. Correct - $3\log n = \log n^3$

2. a. $a = 2$ $d = 3$
   $b = 2$ $O(n^3)$
   $a < b^d$

   b. $a = 1$ $d = 1$
   $b = 10/9$ $O(n)$

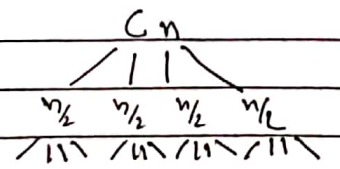
3. $T(n) = T(n/4) + 1$
   $T(n/4) = T(n/4^2) + 1$
   $T(n) = T(n/4^2) + 2$
   $T(n) = T(n/4^k) + k$

   $(n/4^k) = 1$ $\log_4 n = k$

   $T(n) = T(n/2^k) + k$

   $= T(1) + k$
   $O(\log_4 n)$.

4. At level i $n/2^i$ $i = \log_2 n$



$\rightarrow cn$

$\rightarrow 2n$

$\rightarrow 2^2 n$

$- 2^i n.$

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$$T(n) = n(2+2^2+2^3 \ldots 2^i) = \frac{n(2^i-1)}{2}$$

$$O(n^2)$$

5. a. $O(n)$
b. $O(n^2)$
c. $O(n^3)$
d. $\log n$
e. $\log n$
f. $O(n^3)$
g. $O(n^2)$
h. $O(n^{3/2})$
i. $O(n^2)$

6. $T(n) = 2T(n/2 + 17) + n$

$$T(\tfrac{n}{2}+17) = 2T((n/2+17)/2+17) + n/2 + 17.$$

$$T(n) = 2[2T\{(n/2+17)/2 + 17\} + n/2 + 17] + n$$

k^{th} step

$$T(n) = 2^k T[n/2^k + 17 + 17/2 \ldots 17/2^k] + 17k + n + \frac{n}{2} \ldots \frac{n}{2^k}$$

$$\frac{n}{2^k} + 17(1 \ldots 1/2^k) = 1$$

$$\frac{n}{2^k} + 17\left(\frac{1-(1/2)^k}{1/2}\right)$$

$$n + \frac{34(2^k-1)}{2^k} = 1$$

$$n + 34(2^k-1) = 2^k$$

$$n + 34\,2^k - 34 = 2^k$$

$$\log \frac{34-n}{33} = k$$

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$2^{\log_2 \left(\frac{34-n}{33}\right)} T[1] + 17 \log \frac{34-n}{33} + n \left(1 + 1/2 \cdots 1/2^k\right)$

$\log_2 \frac{34-n}{33} T[1] + 17 \log \frac{34-n}{33} + 2n \left(1 - \frac{33}{34-n}\right)$

$O(n \log n)$.

7. a. $3x^2$ & x^2+5.

$\lim_{x \to \infty} \frac{3x^2}{x+5} = \lim_{x \to \infty} \frac{6x}{2x} \quad \lim_{x \to \infty} 3. = 3$

Order of growth is same

8. b. 2 and $\ln x$

$\lim_{x \to \infty} \frac{x}{\ln x} = \lim_{x \to \infty} x = \infty$

Hence growth rate of $x > \ln x$.

8. $5n^2 - 6n = \Theta(n^2)$

$c_1 n^2 < 5n^2 - 6n \leq c_2 n^2$

For several values of c_1 & c_2 This is true so

$5n^2 - 6n = \Theta(n^2)$

9. a. $a = 3 \quad d = 2$

$b = 2 \qquad O(n^2)$

b. $a = 4 \quad d = 2$

$b = 2 \qquad O(n^2 \log n)$.

c. Master Theorem can't be applied here.

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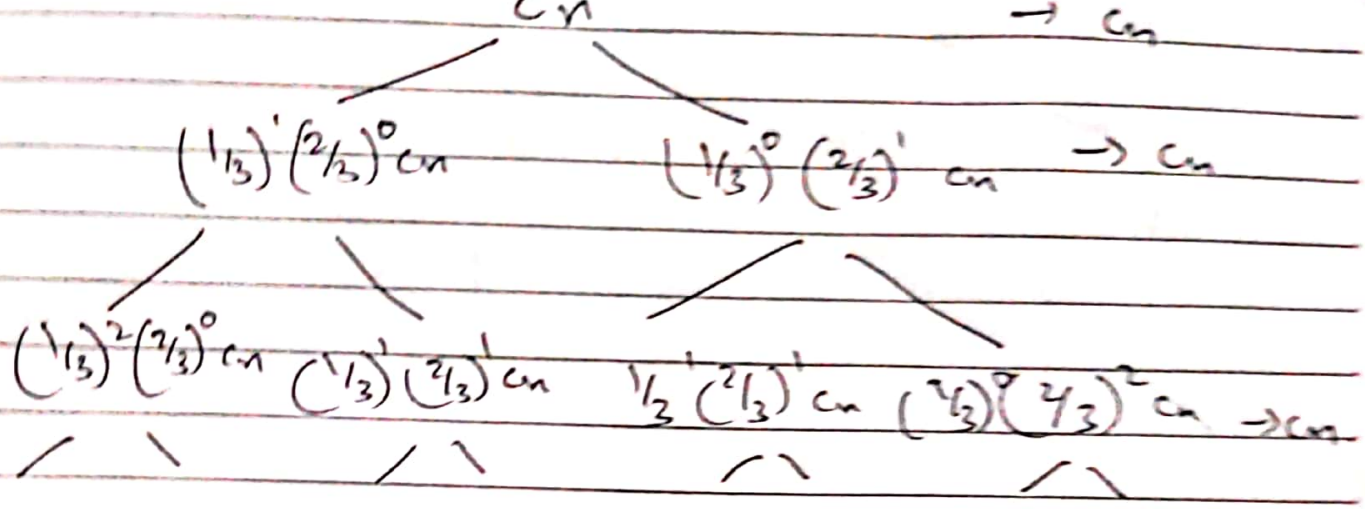
10. $T(n) = T(n/3) + T(2n/3) + cn$



To get $O(1)$ we have to choose the height $\frac{2n}{3}$

At i^{th} level This stops when $(2/3)^i n = 1$ $\Rightarrow$ $n = (3/2)^i$

$i = \log_{3/2} n$.

Each level the cost cn there are $\log_{3/2} n$ levels. Total $cn \log_{3/2} n$

$\therefore \Rightarrow O(n) = O(n \log n)$

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